

$$1.) S = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\vec{p} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$z = ax + by + c$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

$$\leadsto A\vec{p} = z$$

regresia:

$$A^T A \vec{p} = A^T z$$

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$\begin{pmatrix} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{pmatrix}$$

$$\sim \begin{pmatrix} 1 & -4 & -2 & -4 \\ -4 & 1 & -2 & -2 \\ -4 & -4 & -12 & -12 \end{pmatrix} \sim \begin{pmatrix} 4 & -16 & -8 & -16 \\ -4 & 1 & -2 & -2 \\ -4 & -4 & -12 & -12 \end{pmatrix} \sim$$

$$\sim \begin{pmatrix} 1 & -4 & -2 & -4 \\ 0 & -15 & -10 & -18 \\ 0 & -20 & -20 & -28 \end{pmatrix} \sim \begin{pmatrix} 1 & -4 & -2 & -4 \\ 0 & 15 & 10 & 18 \\ 0 & -5 & -5 & -7 \end{pmatrix} \sim \begin{pmatrix} 1 & -4 & -2 & -4 \\ 0 & 15 & 10 & 18 \\ 0 & -15 & -15 & -21 \end{pmatrix} \sim \begin{pmatrix} 1 & -4 & -2 & -4 \\ 0 & 15 & 10 & 18 \\ 0 & 0 & -5 & -3 \end{pmatrix}$$

$$-5c = -3$$

$$c = \frac{3}{5}$$

$$15b = 18 - 10 \cdot \frac{3}{5} = 18 - 6 = 12$$

$$b = \frac{12}{15} = \frac{4}{5}$$

$$a = -4 + \frac{6}{5} + \frac{16}{5} = -\frac{20}{5} + \frac{22}{5} = \frac{2}{5}$$

$$\Pi: z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}$$